

Location and time: Zoom <https://ucsd.zoom.us/j/8314692956> T/Th: 5:00pm-6:20pm

Instructor: Truong Nguyen

E-mail: tqn001@ucsd.edu

Office hours: Tuesdays/Thursdays: 3:45pm-4:45pm and by appointment on zoom
<https://ucsd.zoom.us/j/8314692956>

Course TAs:

Huong Hoang. Email: hlhoang@ucsd.edu

Office hours: Mondays, 3:30pm-4:30pm, <https://ucsd.zoom.us/j/95651285075>
Wednesdays, 1:30pm-2:30pm, : <https://ucsd.zoom.us/j/96535731549>
Fridays, 1:30pm-2:30pm. FAH Collab Space 2301

Xinshuang Liu. Email: xil235@ucsd.edu

Office hours: Tuesdays, 8:30am-10:30am, <https://ucsd.zoom.us/j/93186386079>
Thursdays, 10am-12:00pm. FAH Collab Space 2301

Course Requirements and Grading

Project: Team of 2 to 3 students

- Project Proposal (11/6) 25% Submit by 11:59pm
- Project Update Presentation (11/24, 11/25 11/26) 25% Zoom (7-10pm)
- Final Project Presentation (12/10, 12/11, 12/12) 25% Zoom (7-10pm)
- Project Report (12/14) 25% Submit by 11:59pm
- Zoom link for presentation: <https://ucsd.zoom.us/j/8314692956>

Course grade:

- A+: 97% - 100%
- A: 90% - 96.9%
- A-: 87% - 89.9%
- B+: 83% - 86.9%
- B: 77% - 82.9%
- B-: 74% - 76.9%
- Passing grade for comprehensive exam: B or better

Recommended Textbook:

- *Image Processing for Engineers* by Andrew E. Yagle and Fawwaz T. Ulaby.
Download free copy at <https://ip.eecs.umich.edu/>

- *Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson.*

Recommended Preparation: ECE 109, ECE 153, ECE 161A

Practice homework problems will be handed out periodically. Solutions to these problems will be provided.

Topics:

- Review of DSP, Filtering, Linear-Phase system, DFT
- Overview of Digital Image Processing
- Digital Camera: Image Sensors & Camera Processing Pipeline
- 2D Image & System
- Image Enhancement: Histogram Equalization, Edge Sharpening, Median Filtering, Directional Filtering
- Color Image Processing
- Image Interpolation & Decimation, Multirate DSP
- Wavelets & Multiresolution Analysis
- Image Restoration: Denoising, DeFogging/DeHazing, De-Turbulence
- Image Compression: JPEG & JPEG2000